

Portfolio Choice: Understanding the FOC and Corners

The Problem Setup

You have wealth w_0 to invest. You can:

- Invest amount x in a risky asset
- Keep the rest ($w_0 - x$) safe (earning nothing)

The risky asset:

- Pays off with probability p (you get your money back plus a return)
- Fails with probability $(1 - p)$ (you lose what you invested)

Your utility is CRRA: $u(w) = \frac{w^{1-\sigma}}{1-\sigma}$ where σ is your risk aversion.

Goal: Choose x (how much to invest in risky asset) to maximize expected utility.

What the FOC Tells You

When you take the first-order condition (derivative = 0) and solve, you get some formula for x^* .

The Problem:

The FOC assumes you're at an **interior solution** where $0 < x < w_0$.

But sometimes the "optimal" x from the FOC is:

- Negative (invest less than zero = borrow to invest)
- Bigger than w_0 (invest more than you have)

These violate the constraint that $0 \leq x \leq w_0$!

So after finding x^* from the FOC, you need to **check corners**:

1. Is $x^* \geq 0$? If not, set $x = 0$
2. Is $x^* \leq w_0$? If not, set $x = w_0$

Breaking Down the Passage

Part 1: "FOC is only valid if $\sigma > 0$ "

If $\sigma = 0$ (risk-neutral):

You only care about expected value, not risk. So your decision is simple:

- If the risky asset has positive expected return: invest ALL your wealth ($x = w_0$)
- If the risky asset has negative expected return: invest NOTHING ($x = 0$)

There's no interior solution! You go all-in or stay out completely.

The FOC doesn't apply because the optimal choice is always at a corner.

Why? Risk-neutral people don't care about diversification. They just chase the highest expected return.

Part 2: "FOC is valid provided $p \geq \frac{1}{2}$ "

The FOC gives you a formula for x^* . But we need $x^* \geq 0$ for it to make sense.

When you work out the math (the actual FOC), you find that:

$$x^* \geq 0 \quad \Leftrightarrow \quad p \geq \frac{1}{2}$$

In other words:

- If $p \geq \frac{1}{2}$: The FOC gives a valid interior solution with $0 < x^* < w_0$
- If $p < \frac{1}{2}$: The FOC would give $x^* < 0$ (negative investment), which is impossible!

Part 3: "If $p > \frac{1}{2}$, invest amount strictly positive and strictly below w_0 "

This means:

- The risky asset is "good enough" (succeeds more than half the time)
- You should invest some of your wealth: $x^* > 0$
- But not all of it: $x^* < w_0$ (because you're risk-averse and want to diversify)

Example: If $p = 0.6$ (60% chance of success), $\sigma = 2$, $w_0 = 100$:

The FOC might give something like $x^* = 40$. You invest \$40 in the risky asset and keep \$60 safe.

Part 4: "If $p < \frac{1}{2}$, best option is to invest zero"

If the risky asset fails more than half the time:

The FOC would tell you to invest a negative amount (which means borrow money to short the asset). But you can't do that!

Given the constraint $x \geq 0$, your best option is: **don't invest at all**.

Set $x = 0$ (corner solution).

Intuition: If $p < 0.5$, the risky asset is a bad bet on average. Since you can't short it, you just avoid it completely.

Part 5: "Optimal investment increases in p , decreases in σ "

Increases in p :

- Higher p = risky asset is more likely to pay off
- So you invest more: x^* goes up

Example:

- If $p = 0.6$: maybe you invest \$40
- If $p = 0.7$: you invest \$60 (more confident it will pay off)

Decreases in σ :

- Higher σ = more risk-averse
- So you invest less in the risky asset: x^* goes down

Example:

- If $\sigma = 1$ (moderately risk-averse): invest \$50
- If $\sigma = 3$ (very risk-averse): invest \$20 (prefer safety)

The Big Picture: Checking Corners

Here's the full decision tree:

Step 1: Check if risk-averse

Is $\sigma > 0$?

No ($\sigma = 0$): You're risk-neutral

- Calculate expected return of risky asset
- If positive: $x^* = w_0$ (invest everything)
- If negative: $x^* = 0$ (invest nothing)

Yes ($\sigma > 0$): Go to Step 2

Step 2: Check if risky asset is worth it

Is $p \geq \frac{1}{2}$?

No ($p < \frac{1}{2}$): Risky asset is a bad bet

- Corner solution: $x^* = 0$ (invest nothing)

Yes ($p \geq \frac{1}{2}$): Go to Step 3

Step 3: Use the FOC

Solve the FOC to get x^*

This gives an interior solution: $0 < x^* < w_0$

Check comparative statics:

- Higher $p \Rightarrow$ higher x^* (invest more)
- Higher $\sigma \Rightarrow$ lower x^* (invest less)

Concrete Example

Let's say the risky asset:

- Returns \$2 for every \$1 invested with probability p
- Returns \$0 with probability $(1 - p)$

You have $w_0 = \$100$ and utility $u(w) = \frac{w^{1-\sigma}}{1-\sigma}$.

Case 1: $p = 0.4$ (risky asset fails 60% of the time)

Since $p < 0.5$, the risky asset is a bad bet.

Decision: Don't invest at all. $x^* = 0$.

Keep all \$100 safe.

Case 2: $p = 0.6$, $\sigma = 2$ (60% success, moderately risk-averse)

Since $p > 0.5$ and $\sigma > 0$, use the FOC.

The FOC would give some formula like:

$$x^* = f(p, \sigma, w_0)$$

Let's say it gives $x^* = \$40$.

Decision: Invest \$40 in risky asset, keep \$60 safe.

Expected outcome:

- 60% chance: risky asset pays \$80, plus \$60 safe = \$140 total
- 40% chance: risky asset pays \$0, plus \$60 safe = \$60 total

Case 3: Same but $\sigma = 4$ (very risk-averse)

FOC might now give $x^* = \$20$.

Decision: Invest only \$20 in risky asset (less than before because you're more scared of risk).

Summary

Key Takeaways:

1. **FOC only works if $\sigma > 0$ (risk-averse)**
 - If risk-neutral ($\sigma = 0$): go all-in or stay out
2. **FOC gives valid x^* only if $p \geq \frac{1}{2}$**
 - If $p < \frac{1}{2}$: corner solution, set $x = 0$
3. **Interior solution when $p > \frac{1}{2}$ and $\sigma > 0$**
 - Invest some but not all: $0 < x^* < w_0$
4. **Comparative statics:**
 - x^* increases in p (better odds $\Rightarrow$ invest more)
 - x^* decreases in σ (more risk-averse $\Rightarrow$ invest less)
5. **Always check corners!**
 - Is $x^* \geq 0$? If not, set $x = 0$
 - Is $x^* \leq w_0$? If not, set $x = w_0$